

Opinion Network Formation from a Class Survey

Assignment 1: Distributed Protocols for Computer Networks

18 September 2026

Team Name

Autobots

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GitHub Link to Our Code

https://github.com/Devansh-236/DPCN_A1_Autobots

The repository contains the complete, reproducible pipeline. `python src/run_all.py` regenerates every number and every figure in this report from the raw CSV in one command.

Nothing here was typed by hand: each stage writes its numbers to `results/*.json` and the figures read those files, so text, tables and charts cannot drift apart. All randomness is seeded.

Dataset Documentation

What the raw file contains

The export `Survey_Results_UC.csv` holds **96 response rows** and **61 columns**: one response ID plus **60 opinion statements**. The statements are organised into four thematic blocks of fifteen, identifiable from their column prefixes:

Code	Domain	Items	A statement from the block
T	Technology	15	<i>“AI will improve society more than it will create problems.”</i>
E	Education	15	<i>“Project-based learning develops deeper understanding than exams.”</i>
S	Ethics & Society	15	<i>“Diverse teams generally make better decisions.”</i>
V	Environment	15	<i>“Climate change requires immediate global action.”</i>

Every cell is one of seven strings. Five are Likert points; two are non-answers.

Interpretation decisions

1. Likert points become a symmetric integer scale. Strongly Disagree $\rightarrow -2$, Disagree $\rightarrow -1$, Neutral $\rightarrow 0$, Agree $\rightarrow +1$, Strongly Agree $\rightarrow +2$. The scale is centred on Neutral so that “no opinion” is numerically zero and disagreement is genuinely the negative of agreement. This matters because every similarity we compute is an inner product of these vectors.

2. “No Comments” is treated as missing, not as Neutral. The survey offered *Neutral* and *No Comments* as separate choices, so a respondent who picked *No Comments* (37 cells) deliberately declined rather than taking

a middle position. Coding it as 0 would manufacture agreement between everyone who skipped a question. It is coded as missing instead.

3. Blank cells (505, or 8.8% of the 5,760-cell grid) are missing.

4. Non-participants are removed, partial participants are kept. Missingness is not scattered. It is *block-wise*, exactly what one sees when respondents abandon a form (Figure 1, right). Five respondents answered nothing at all, four stopped after the 15-item Technology block, one stopped after 45 items and one after 50. We therefore drop any respondent who answered fewer than **50% of the 60 items**, which removes exactly the nine people who never really participated (IDs 44, 60, 68, 73, 78, 30, 39, 77, 87) and keeps everyone who worked through at least three blocks. This leaves **87 respondents**, our network’s nodes, with a residual missing rate of only **1.19%** (62 cells). Those 62 cells are never imputed: every similarity is computed on the items a given *pair* both answered (pairwise-complete), with a minimum overlap of 20 items.

What the cleaned data looks like

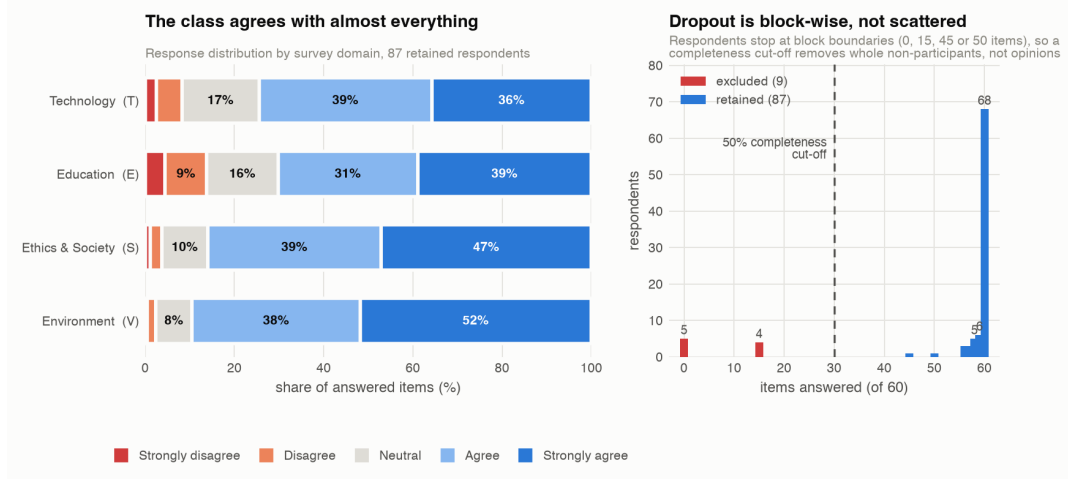


Figure 1: The cleaned response matrix. Left: agreement dominates every domain, and the Environment block is the most consensual. Right: dropout happens at block boundaries, which is why a completeness cut-off removes non-participants rather than opinions.

After cleaning, **80.0%** of all answers are *Agree* or *Strongly Agree*, 12.9% are *Neutral*, and only **7.1%** are any form of disagreement. The grand mean is **+1.14** on a scale running to +2. Fifty-eight of the 60 statements have a positive mean; only two are net-negative.

This single fact drives the entire methodology that follows. **A class this agreeable cannot be networked on raw agreement.** If we connected respondents whose answers simply look alike, we would connect everyone to everyone, and the network would describe the questionnaire rather than the people answering it. The next section addresses this directly.

Pipeline Followed

The network we build

Nodes are the 87 retained respondents. **Edges** are undirected and weighted: an edge (u, v) exists when u and v agree with each other *more than the class as a whole agrees about anything*, and its weight is how much more. The network is therefore a map of **residual opinion similarity**: who shares an outlook once the shared class consensus has been subtracted.

Step 1. Encode and clean

As documented above: map to $\{-2, \dots, +2\}$, treat both non-answer tokens as missing, drop the nine non-participants. Output: an 87×60 matrix X with NaNs.

Step 2. Remove the class consensus

Each item j is standardised across respondents,

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad \mu_j, \sigma_j \text{ computed over the respondents who answered item } j.$$

μ_j is the class consensus on item j ; z_{ij} is respondent i 's **deviation** from it. Dividing by σ_j gives every item equal say, so that a near-unanimous item (V15, sd 0.55) cannot outvote a genuinely contested one (E04, sd 1.16).

Figure 2 (left) shows why this step is not cosmetic. Before standardisation, **91.2%** of all 3,741 respondent pairs have a *positive* correlation (mean +0.264): everybody endorses climate action and everybody is cool on written exams, so everybody looks alike. After standardisation the distribution is centred on zero (mean -0.011), and similarity finally means something specific.

Step 3. Measure pairwise similarity

For each pair we take the Pearson correlation of their standardised vectors over the items they both answered:

$$S_{uv} = \text{corr}(z_{u, \mathcal{J}_{uv}}, z_{v, \mathcal{J}_{uv}}), \quad \mathcal{J}_{uv} = \{j : \text{both answered } j\}, \quad |\mathcal{J}_{uv}| \geq 20.$$

Pearson (rather than cosine) re-centres each respondent, which removes any residual *acquiescence bias*, so a person who agrees enthusiastically with everything no longer looks similar to every other enthusiast purely on that basis.

Step 4. Decide which similarities are real

A threshold picked by eye is the weakest point of most similarity networks, so we derive one from a **null model**. We permute each item independently across respondents: this preserves every item's marginal distribution exactly while destroying any genuine respondent-to-respondent coupling. Re-running steps 2–3 on 60 such permutations yields **224,460 null pair similarities** (mean -0.012 , sd 0.130).

We keep an edge when its similarity exceeds the **99th percentile of that null**, $\tau = 0.299$. This gives **88 edges** where 37 would be expected by chance alone, a 2.4-fold enrichment. Figure 2 (right) overlays the observed and null distributions.

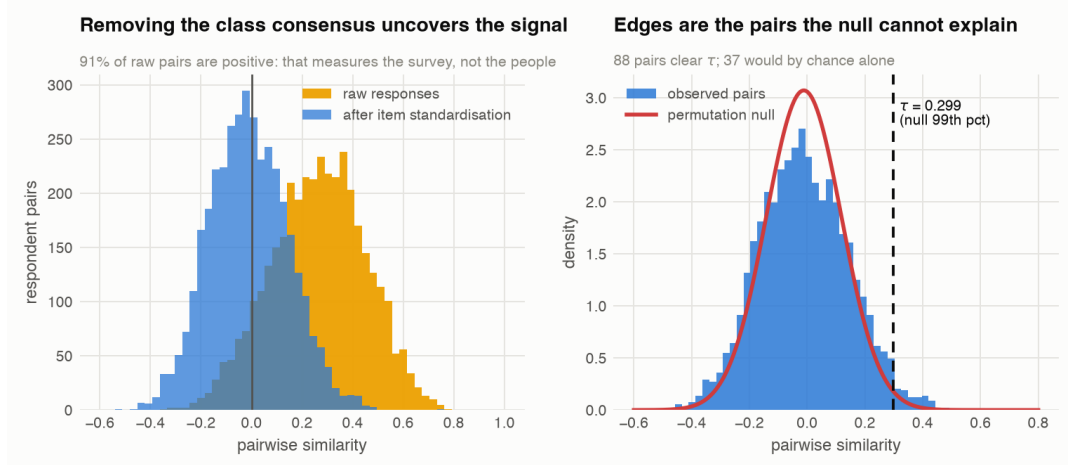


Figure 2: Left: subtracting the class consensus moves the similarity distribution from “almost everything is positive” to centred on zero. Right: the edge threshold is the 99th percentile of a permutation null, not a hand-picked number.

Because any threshold is a judgement call, we swept it from 0.05 to 0.525 and recorded how the network responds (Figure 3). The sweep shows a clear regime change: below ≈ 0.25 the graph is a single connected blob of 200+ edges with nothing to say; above ≈ 0.40 it shatters into dust. Our null-derived $\tau = 0.299$ lands inside the informative band, just past the point where the giant component detaches from the periphery.

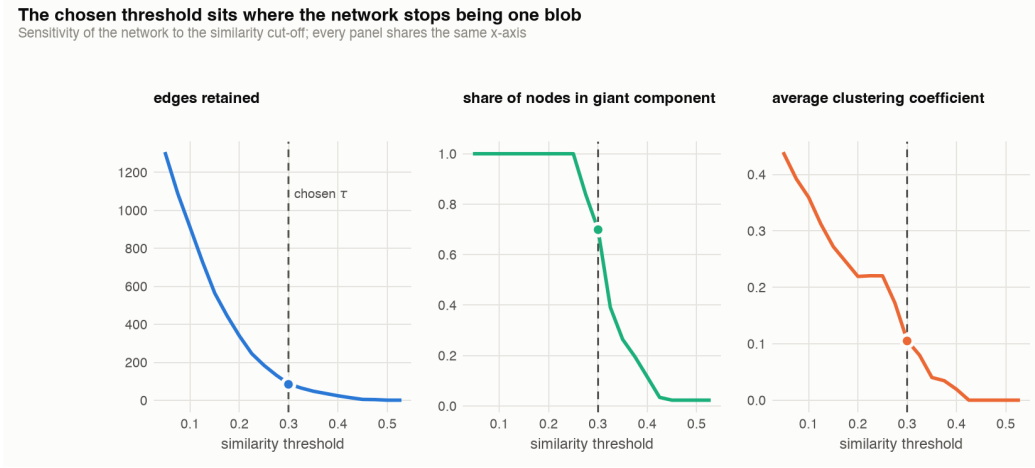


Figure 3: Threshold sensitivity. The null-derived τ sits in the informative regime between “one blob” and “no network”.

Step 5. Analyse

We then compute global structure, centralities, and communities (Louvain on the weighted graph, best of 50 random restarts), compare against 200 Erdős–Rényi and 200 configuration-model graphs, and, as a robustness check on every structural claim, rebuild the whole network a second way, as a **mutual 5-nearest-neighbour graph** (156 edges, connected by construction, no threshold at all).

Analysis and Visualizations

The network

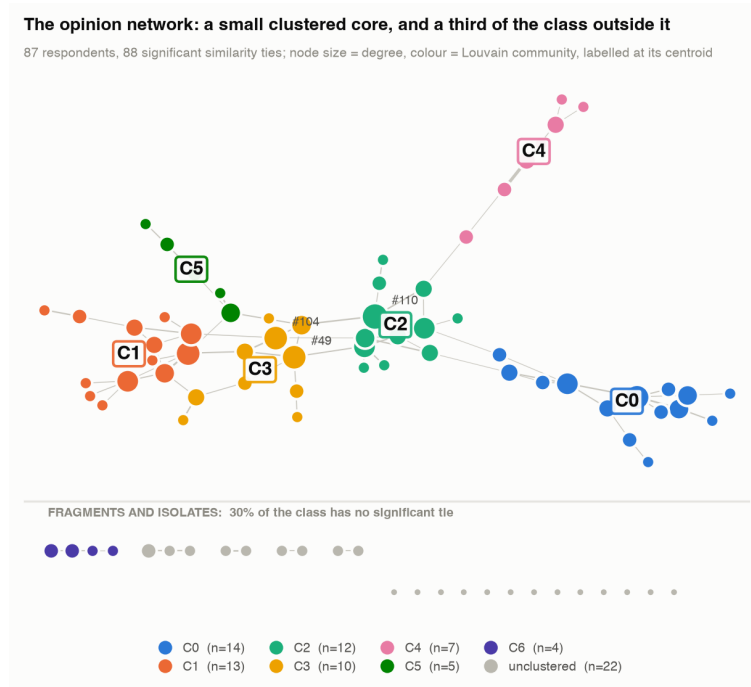


Figure 4: The opinion network. Seven Louvain communities occupy the 61-node giant component; the remaining 26 respondents form small fragments or sit alone.

Global metrics

Metric	Opinion network G	Mutual-5NN check	Erdős–Rényi	Configuration model
Nodes	87	87	87	87
Edges	88	156	88	88
Density	0.0235	0.0417	0.0235	0.0235
Average degree	2.02	3.59	2.02	2.02
Max / median degree	7 / 1	5 / 4	n/a	n/a
Isolates	13	0	n/a	n/a
Connected components	19	1	n/a	n/a
Giant component	61 (70.1%)	87 (100%)	n/a	n/a
Average clustering C	0.1009	0.1502	0.0154	0.0168
Transitivity	0.1806	0.1623	n/a	n/a
Avg. path length L (giant)	5.37	4.03	5.30	4.63
Diameter (giant)	12	8	n/a	n/a
Degree assortativity	+0.125	+0.140	~ 0	~ 0
Mean edge weight	0.371	0.322	n/a	n/a

The headline structural result is the **small-world signature**: clustering is $C/C_{\text{rand}} = 6.6\times$ the value for random graphs matched on both n and m (the analytic Erdős–Rényi prediction $C = p$ gives $4.3\times$; see below), while the average path length is $L/L_{\text{rand}} = 1.01\times$ it, giving $\sigma = 6.5$. Opinion ties are strongly triadic: if you agree with two people, those two tend to agree with each other, yet the graph is no harder to traverse than a random one of the same density.

Degree is also over-dispersed relative to the Poisson expectation (Figure 5, left): more isolates *and* more high-degree hubs than chance allows. The compact way to say this is the ratio $\text{Var}(k)/\langle k \rangle$, which equals exactly 1 for a Poisson degree sequence. Ours is **1.43**, mildly over-dispersed, but nowhere near the fat-tailed regime where that ratio runs into the hundreds. With degrees spanning only 0 to 7 there is no dynamic range for a power law, which is why Figure 5 bins linearly; logarithmic bins normalised by bin width are the right instrument only when the degree axis covers orders of magnitude, and here it covers less than one.

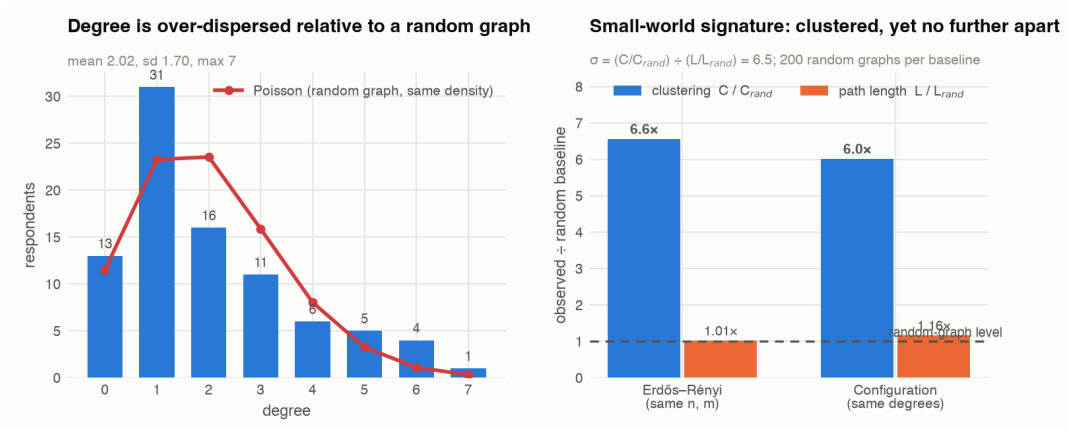


Figure 5: Left: the observed degree distribution against the Poisson curve for a random graph of the same density. Right: both structural quantities expressed as a ratio to their random-graph expectation, which is the small-world signature.

How the network compares with a random graph

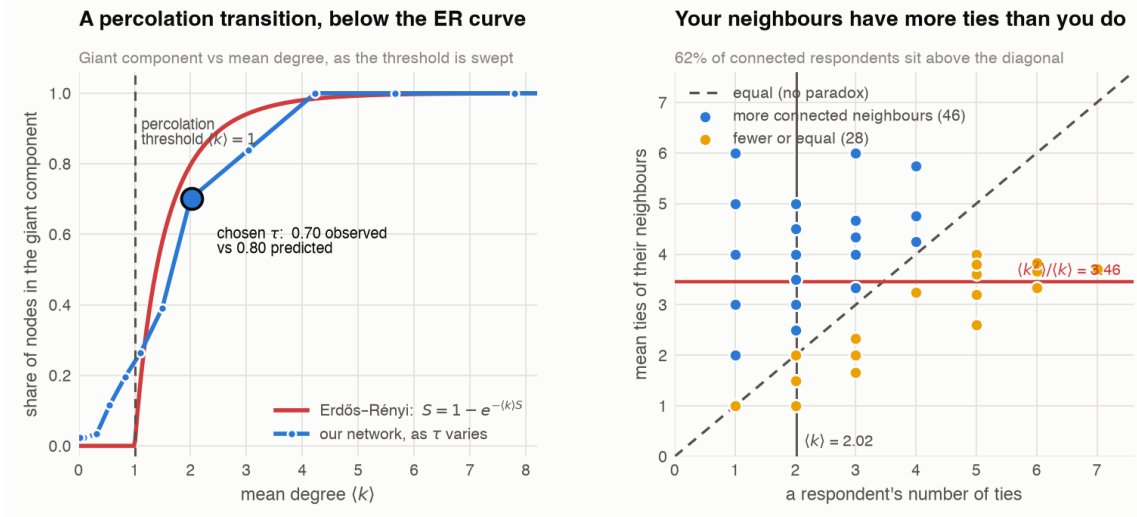


Figure 6: Left: sweeping the threshold traces out a percolation transition, plotted against the Erdős-Rényi prediction. Right: almost two thirds of connected respondents have fewer ties than their own neighbours average.

The threshold sweep of Figure 3 is really a percolation experiment: every threshold yields a network with some mean degree, and lowering τ raises $\langle k \rangle$ continuously. For an Erdős-Rényi graph the fraction S of nodes in the giant component solves

$$S = 1 - e^{-\langle k \rangle S},$$

which has only the trivial root $S = 0$ until $\langle k \rangle = 1$ and a positive root above it. At our threshold $\langle k \rangle = 2.02$: past the percolation point, but well short of the connectivity threshold $\langle k \rangle = \ln N = 4.47$. That is the **supercritical regime**, a giant component coexisting with small tree-like fragments and isolated nodes, and it is exactly what Figure 4 shows: 61 nodes in the giant, 5 small fragments, 13 isolates.

The network sits *below* the random-graph curve, though: 70.1% of nodes in the giant against a predicted 80.3%. That deficit is what clustering costs. Our 88 edges are partly spent closing triangles inside communities instead of reaching new people. Below $\langle k \rangle = 1$ the observed curve sits *above* the ER line for the mirror-image reason: the very strongest similarity ties are not scattered at random but concentrated among a few mutually similar respondents.

Clustering. For an Erdős-Rényi graph $C = p$ exactly, and here $p = 0.0235$ against a measured $C = 0.1009$, a factor of **4.3**. (The $6.6\times$ quoted above compares instead against simulated graphs matched on n and m , whose average clustering is depressed by nodes of degree below 2 contributing zero. Both are defensible; the analytic figure is the more conservative.)

Path length. The Erdős-Rényi estimate $\langle d \rangle \approx \ln N / \ln \langle k \rangle$ gives **4.26** for the 61-node giant component at $\langle k \rangle = 2.62$, against a measured **5.37**, longer than random, for the same reason clustering is higher. But measured against structured topologies of the same size it is small: an open chain would give $(n + 1)/3 = 20.7$, a degree-4 ring lattice $\approx N/8 = 7.6$, a two-dimensional lattice $\approx \sqrt{N} = 7.8$. Opinion similarity does not lay respondents out along a line or across a grid.

Neighbours are better connected than you are. Reaching a node by following an edge is size-biased sampling: a respondent with k ties is k times more likely to be arrived at than one with a single tie. The mean degree of a neighbour is therefore

$$\langle k_{nn} \rangle = \frac{\langle k^2 \rangle}{\langle k \rangle} = \langle k \rangle + \frac{\text{Var}(k)}{\langle k \rangle} = 2.02 + 1.43 = 3.46,$$

and because $\text{Var}(k) > 0$ for any network whose nodes are not all of equal degree, the inequality $\langle k_{nn} \rangle > \langle k \rangle$ holds universally. Walking the edge list directly gives 3.29, and **62% of connected respondents have fewer ties than their neighbours average** (Figure 6, right). Note that the excess, 1.43, is the same degree-dispersion ratio as above; an Erdős-Rényi graph, where $\text{Var}(k) = \langle k \rangle$, would give exactly one extra tie. The effect is a property of how edges sample nodes, not of how opinions form.

Central respondents

Respondent	Degree	Strength	Betweenness	Closeness	Eigenvector
#110	7	2.70	0.150	0.197	0.379
#104	6	2.27	0.111	0.186	0.359
#49	6	2.27	0.095	0.185	0.357
#119	6	2.39	0.090	0.129	0.022
#95	6	2.25	0.093	0.160	0.201

The five centrality measures agree closely (Spearman ρ of degree with strength 0.97, with betweenness 0.89, with closeness 0.78, with eigenvector 0.74), so there is a single, unambiguous core rather than competing notions of importance. Respondent **#110** is the class’s “median voice”: the person whose opinion profile is closest to the largest number of classmates. Note that **#119** has high degree but near-zero eigenvector centrality, so it is a hub of an outlying branch, not of the core.

Communities

Louvain finds **six communities** partitioning the 61-node giant component, with modularity $Q = 0.677$ measured on that component, plus a seventh group (**C6**, four people) that is a connected component entirely of its own. The partition is highly stable: across 50 random restarts the mean pairwise adjusted Rand index is **0.94**. The blocks are visible directly in the similarity matrix (Figure 7).

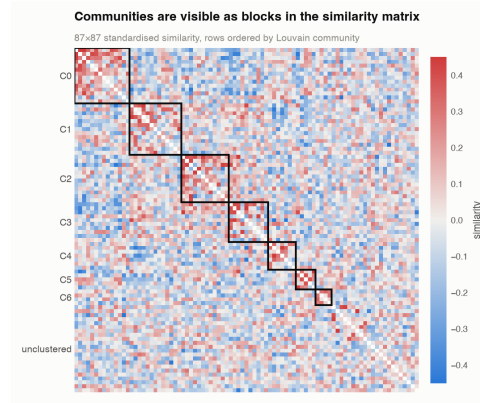


Figure 7: The 87×87 similarity matrix with rows ordered by community. Red diagonal blocks are the communities; the unclustered periphery shows no block structure.

Homophily is real but moderate: mean similarity **within** a community is +0.108 against **−0.029 between** communities, a gap of 0.137 (Cohen’s $d = 0.78$).

#	Size	Mean score	z_T	z_E	z_S	z_V
C0	14	0.85	−0.10	−0.30	−0.47	− 0.75
C1	13	1.20	+0.15	+0.06	+0.01	+0.11
C2	12	1.39	+0.01	+0.34	+0.46	+0.45
C3	10	1.37	+0.29	+0.10	+0.43	+0.38
C4	7	1.02	+0.07	−0.33	−0.18	−0.20
C5	5	1.29	−0.11	+0.02	+0.37	+0.47
C6	4	0.78	− 0.60	−0.10	−0.39	−0.68

Read against the class mean, each is a coherent stance. **C0**, the *reserved* group, is uniformly restrained and coolest on environmental commitment (V05 $z = -1.19$). **C1**, *campus-first technologists*, sit near the class average but back AI education (T04 +0.66) while rejecting online learning (E04 −0.77). **C2** are *research-minded reformers* (E06 +0.90) who are notably cool on AI optimism (T01 −0.55). **C3**, the *regulation-minded progressives*, pair the strongest enthusiasm for technology with the strongest demand for regulating it (T12 +0.74, T10 +0.70). **C4**, *exam-hall traditionalists*, defend examinations and attendance while rejecting project-based learning outright (E01 −1.62). **C5**, *techno-optimist greens*, back environmental goals but dismisses data-protection-first thinking (T10 −1.58). **C6**, four *contrarians* forming their own component, reject the platform-harm narrative entirely (T14 −2.26, S06 −1.72).

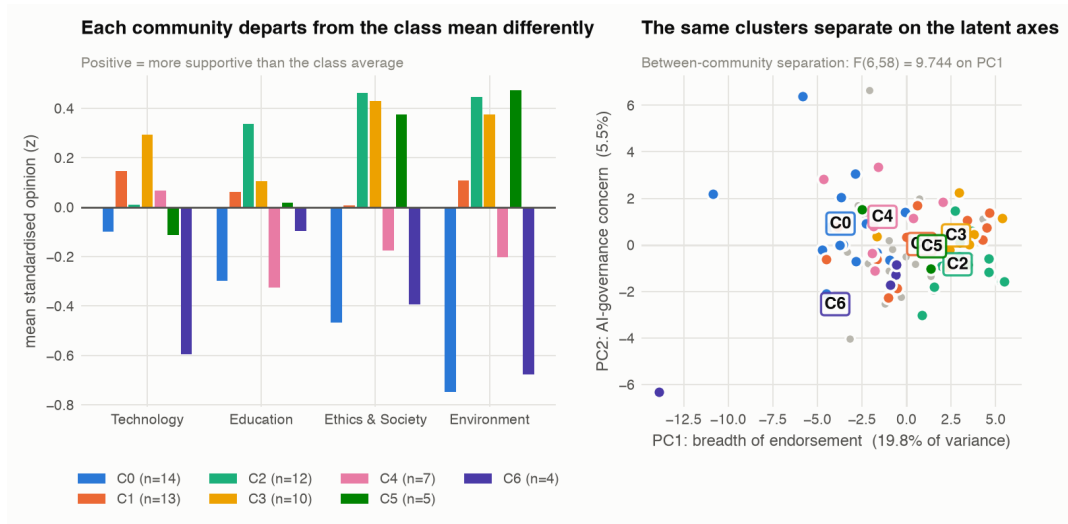


Figure 8: Left: how far each community departs from the class mean, by domain. Right: the same communities plotted on the two dominant latent opinion axes.

The communities are not an artefact of the clustering algorithm. Projecting respondents onto the principal axes of the opinion matrix (Figure 8, right) separates them significantly: $F(6, 58) = 9.74$ on PC1 and $F(6, 58) = 4.15$ on PC2. **PC1 (19.8% of variance)** is a *breadth of endorsement* axis, loading on S09 (inclusive debate), V09/V10 (reuse and conscious consumption) and E10 (innovation over rote), opposed by E03 (compulsory attendance) and E02 (exams measure knowledge). **PC2 (5.5%)** is an *AI-governance concern* axis: T12, T14, T10. PC1 correlates with eigenvector centrality at $\rho = +0.47$: the network's core is made of broad endorsers, and scepticism pushes a respondent to the periphery.

Where the class actually divides

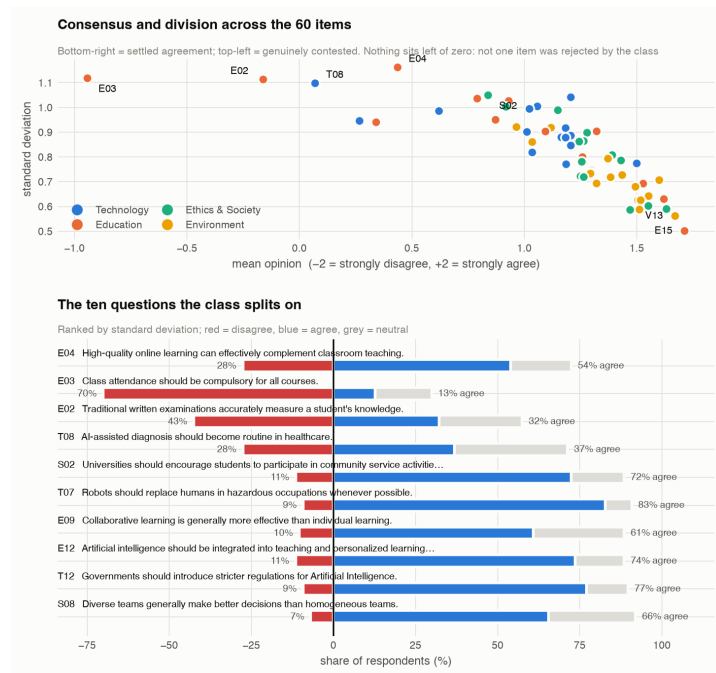


Figure 9: Top: every item positioned by mean opinion and disagreement. Bottom: the ten most contested statements.

Disagreement is concentrated in **Education**: the three most contested items of all are E-block statements, and the Education block has the second-highest mean item sd (0.881) behind Technology (0.916), against Environment's 0.719. Only two statements in the whole survey have a negative mean these are E03 (compulsory attendance, -0.94 , rejected 70:13) and E02 (written exams measure knowledge, -0.16 , rejected 43:32). A third, T08 (routine AI diagnosis), sits essentially at zero ($+0.07$) with the class split 28:37.

Do the survey's four themes exist in the data?

We built a second network with the 60 **items** as nodes, joining two items when responses to them co-vary in the top decile of all item pairs. If the survey's T/E/S/V design reflected how students actually think, the detected clusters should reproduce it.

They do not. Five clusters emerge (sizes 14, 14, 13, 4, 3) and every one of them mixes themes; agreement with the survey's own domains is **ARI = 0.12**, barely above the 0 of an unrelated partition.

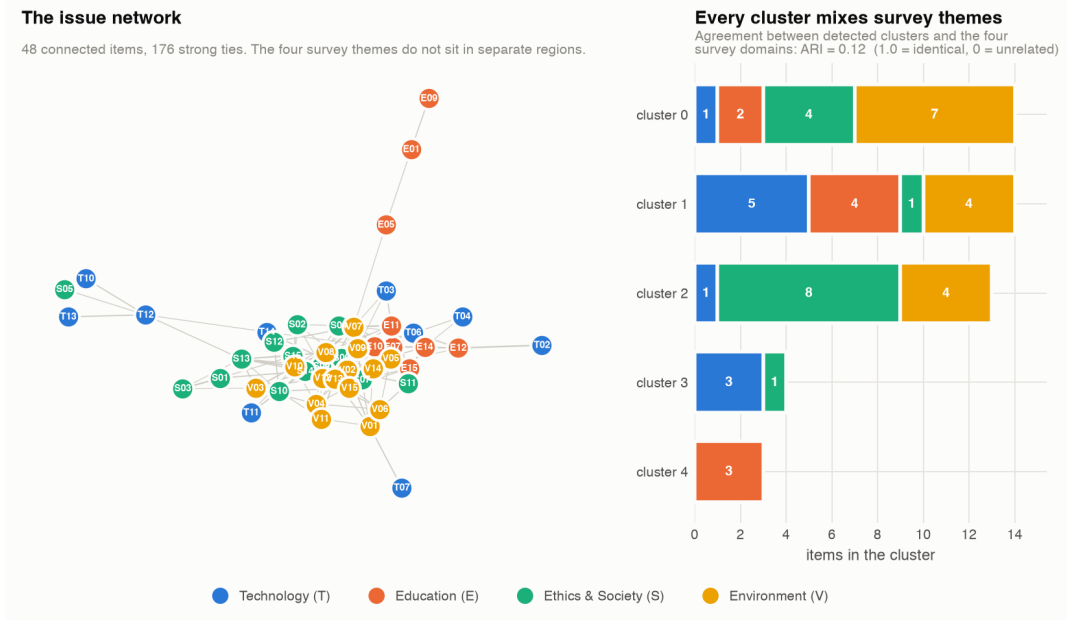


Figure 10: Left: the issue network. Right: what each detected cluster is made of, showing that no cluster corresponds to a survey theme.

Results and Discussion

1. The class is overwhelmingly consensual, and that is the primary finding. Eighty percent of all answers are agreement and the grand mean is +1.14. On 57 of 60 statements the class agrees, on average, with the proposition. There is no polarisation here in the political-science sense: there is a broad shared value system (sustainability, ethics, inclusion, lifelong learning) and the network structure lives in the thin residual layer above it.

2. Once consensus is removed, genuine but weak structure remains. Observed pair similarities have sd 0.155 against the null's 0.130. Since $0.155^2 - 0.130^2 = 0.0071$, the *true* similarity between two classmates has a standard deviation of only about **0.084**: real, reproducible, and small. Any honest reading of this network must start from that number: 88 edges is $2.4\times$ what chance gives, but it is not a densely woven social fabric. The 60-item survey simply does not have enough resolution to measure agreement between two individuals precisely, which is exactly why we threshold against a null rather than trusting individual pair values.

3. The network is a small world with a large periphery. Clustering is $6.6\times$ random while path length is unchanged ($\sigma = 6.5$), and degree assortativity is positive (+0.125), so well-connected respondents connect to each other, forming the core visible in Figure 4. But **13 respondents have no significant tie at all** and 26 (30% of the class) sit outside the giant component. These are not people with no opinions; they are people whose *combination* of opinions is unusual enough that nobody else in the class shares it. In an opinion-dynamics reading, they are the individuals least exposed to reinforcement and most likely to shift. The Laplacian records the same fact spectrally: it has exactly 19 zero eigenvalues, one per component, and the giant component's algebraic connectivity is only $\lambda_2 = 0.016$, so even the connected core is barely held together, and anything diffusing across it would equilibrate slowly.

4. The people you agree with agree with more people than you do. The mean degree of a neighbour is $\langle k_{nn} \rangle = 3.46$ against $\langle k \rangle = 2.02$, and 62% of connected respondents sit below their own neighbours' average. This is the familiar sampling effect, since popular nodes are over-represented among everybody's neighbours, but in an opinion network it carries a specific reading: the classmates whose views you share are typically people whose views are *also* shared by others. Opinions close to the consensus are over-represented among anyone's neighbours. That fits finding 2's observation that PC1 correlates with eigenvector centrality at $\rho = +0.47$: a core of broad endorsers that everyone connects into, and a periphery whose unusual combinations connect to nobody.

5. Seven stable communities, differing in degree rather than in direction. $Q = 0.677$ with a 94% stable partition across restarts, and within-community similarity exceeds between-community similarity by 0.137 ($d = 0.78$). Crucially, Figure 8 shows the communities separating mostly along *how strongly* they endorse the shared value system (PC1, 19.8% of variance), not along opposing camps. The exceptions are the interesting ones: **C4** rejects project-based and collaborative learning while defending exams and attendance, a coherent traditionalist position; **C6**, four respondents, rejects the entire platform-harm narrative (T14 at $z = -2.26$); **C3** pairs enthusiasm for technology with the strongest demand for regulating it; and **C5** takes the opposite tack, backing environmental goals while dismissing data-protection-first thinking. These are recognisable, internally consistent worldviews, not noise.

6. Education is where the class genuinely argues. The contested items are pedagogical, not ethical: compulsory attendance (70% against), the validity of written examinations (43% against), online learning (28/54 split), and AI-assisted medical diagnosis (28/37 split). Students hold a shared ethical and environmental outlook but disagree sharply about how they should be taught, and the traditionalist/reformist split on that question is the single strongest axis in the data.

7. The survey’s thematic blocks are not how students actually organise their opinions. With $ARI = 0.12$, the empirically detected issue clusters cut across Technology / Education / Ethics / Environment. The largest cluster joins seven Environment items with four Ethics items, two Education items and one Technology item, a “collective responsibility” dimension that the survey’s own structure splits across three sections. Opinion is organised by underlying value, not by topic heading.

Robustness

Every structural claim survives rebuilding the network a completely different way. The mutual-5NN graph, which uses no threshold and no null model and is connected by construction, reproduces the elevated clustering (0.150) and the positive assortativity (+0.140), and gives a community partition agreeing with the main one at $ARI = 0.42$ despite having 10 communities instead of 7. The threshold sweep also shows the clustering excess is not an artefact of τ : measured against random graphs of matching size at each cut-off, it is present everywhere and grows monotonically: $1.5\times$ at $\tau = 0.10$, $2.4\times$ at 0.20, $5.1\times$ at 0.25, $7.7\times$ at 0.30 and $10.4\times$ at 0.40. Loosening the threshold weakens the result but never reverses it.

Limitations

Sixty items is few for estimating a correlation between two individuals, and the null sd of 0.130 is essentially $1/\sqrt{60}$, and this measurement noise, not methodology, is what caps the density of the network. *The response scale is severely skewed*, so the effective dynamic range is two points (Agree / Strongly Agree) rather than five. *Edges represent opinion similarity, not social contact*: this network says who thinks alike, and nothing about who talks to whom, so it cannot by itself distinguish homophily from social influence. *A 42% false-discovery rate* (37 of 88 edges expected by chance) means individual edges should not be over-interpreted; only aggregate structure is trustworthy, which is why every conclusion above rests on distributions and communities rather than on particular ties.

Individual Contribution

Member	Tasks completed
Devansh Varshney	Data and item analysis. Likert encoding, the “No Comments” decision and the completeness cut (<code>prepare_data.py</code>); item consensus and polarisation statistics; Figures 1, 9 and 10; Dataset Documentation section.
Vansh Agarwal	Network construction. Item standardisation, similarity, the permutation null, thresholding and the sensitivity sweep (<code>build_network.py</code>); the mutual-kNN check; Figures 2, 3 and 6; Pipeline Followed section.
Aasrith Reddy Vedanaparti	Network analysis. Metrics, the five centralities, communities, random-graph and theoretical benchmarks (<code>analyze.py</code>); latent axes; Figures 4, 5, 7 and 8; Analysis and Results sections.

Each contribution carries equal weight: one pipeline stage, one analysis area, three or four figures and two report sections. All three jointly agreed the methodology and proof-read the report.